

WASP-39b

R-1493

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_180524 · created 2026-08-01T14:01:37+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458262.791 +/- 0.007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.023
Transit depth (Rp/Rs)^2	3.54 +/- 0.88 [%]
Orbital Inclination (inc)	85.24 +/- 0.81
Airmass coefficient 1 (a1)	1.097 +/- 0.015
Airmass coefficient 2 (a2)	-0.07 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	None
Optimal Aperture	4.78
Optimal Annulus	15.97
Transit Duration (day)	0.075 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.188 ± 0.023 vs 0.1457 (deviation 1.56σ)
Duration fit vs published	0.075 d vs 0.12 d (deviation 1.23σ)
Mid-time O–C	28.3 min
Depth SNR	6.9
Residual scatter	1.36 %

Light curve

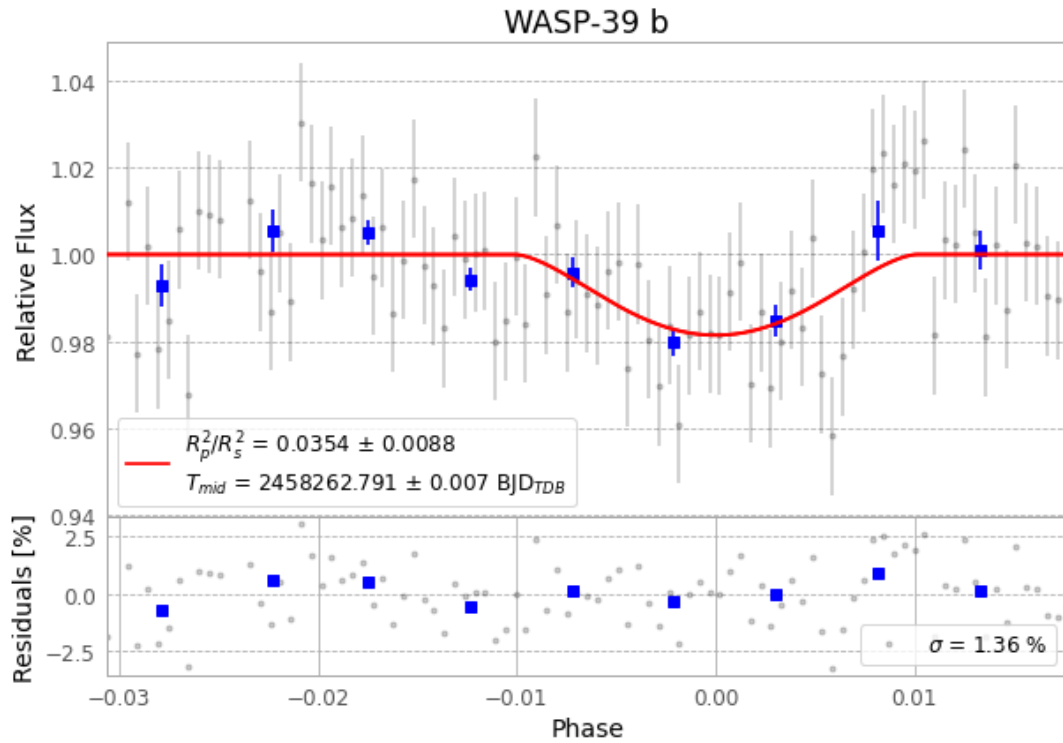


Figure 1 — FinalLightCurve_WASP-39 b_2018-05-23.png (SHA-256 6d838d01c5cf1cc4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 216], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 f1f6761ffa89b0ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

94 FITS frames (62 MB), WASP-39180524035115.FITS → WASP-39180524083312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-180524.log (6ebab134c1af...), FinalLightCurve_WASP-39 b_2018-05-23.png (6d838d01c5cf...), FinalLightCurve_WASP-39 b_2018-05-23.csv (b0f7cc1fdaef...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 14:01Z.